

Item description	Manufacturer	Part number	Price*
 Calipers. Callipers made of ESD-safe material; non-metallic high-tech material with 60 per cent fibre-glass content; surface resistance 10^9 ohms to 10^9 ohms	WIHA	4112108	₹ 4864
 Fume extractors. Remove noxious fumes quickly, safely and efficiently with maximum absorption capacity of 1m ³ /minute	DURATOOL	D00681	₹ 3701
 Chemical dispensers. Average surface resistivity of 10^9 - 10^{10} Ω per square. Do not react with solvents like TCE or alcohol	MULTICOMP	146-0011	₹ 1998
 Writing instruments. Made from dissipative thermoplastic material for use in ESD areas	CK TOOLS	Z1123	₹ 390

*Prices of tools are as listed on element14.com website

Surface resistivity plays a key role in monitoring and examining the resistive characteristics of packaging materials, work bench surfaces, floor mats, storage bags and other items used in the electronics assembly and packaging industry.

Handheld and work-holding tools. Brushes as an anti-static tool enable manufacturers of electronic and static-charge-sensitive products to clean and make products that do not create static charges. Some vendors also provide custom-made clean room/ESD brushes.

Cutters with dissipative, soft-grip handles and induction-hardened cutting edges are used for superior performance. Pliers are used for gripping, holding and bending, and any electrostatic energy present is discharged through the handles in

a gradual and controlled manner, which protects components endangered by electrostatic discharge. Ergonomically-designed cutters and pliers with a good hand grip are especially important for professional technicians for lead cutting.

Another work-holding tool, a tweezer, is used for holding, pulling, squeezing, picking up components and plucking jobs. The protective coating on these prevents harm to sensitive electronics from ESD.

Magnifying lamps provide full protection against ESD. The complete lamp (head, lens, joint and arm) is grounded and has special static-dissipative materials to smoothly take away static charges in a controlled manner, preventing any build up of static electricity, which gives complete protection against ESD.

Ionisers. In order to reduce potential damage that could be caused by ESD, ionisers are another extensively used equipment. Ionisers are most commonly used by manufacturers of electronic and surface-mount devices (SMDs) as well as printed circuit board (PCB) industries.

Sometimes, static-control products such as ESD flooring, mats and wrist straps are unable to completely eliminate static charges. An ioniser removes static charges from surfaces, thereby providing additional protection. Depending on the application usage and industry, new-age ionisers are available in benchtop, gun, nozzle, bar and overhead box form factors.

Anti-static wristbands. ESD wrist straps or wristbands allow the personnel doing assembly, maintenance or repair work to work without being concerned about static electricity. Fitted around the wrist, as the name suggests, these are usually fully-adjustable and dissipate potential harmful charges to the ground, thereby allowing personnel to handle ESD-sensitive devices safely.

Today's wrist straps are available with and without cords. Wired wrist

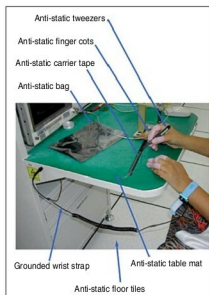


Fig. 3: An example of a static-safe work bench (Image courtesy: www.minicircuits.com)

Did you know

Many objects in today's environment can store thousands of volts in electrostatic charges. Yet, it only takes 25 electrostatic volts to irreparably damage an integrated circuit.

—Data courtesy: *Electrical Construction & Maintenance (EC&M)*

How to identify electrostatic discharge sensitive devices

The labels shown here are most frequently used on packaging materials, holders and cases to notify anyone handling ESD devices. This alerts the user to utilise static-safe procedures before handling such devices. The one on the left is preferred and highly used. At times, ESD devices also come with only a cautionary note that reads, "Contains parts and assemblies susceptible to damage by ESD."

